Day 6 — HTML Fundamentals

Detailed Answers, Complete Programs, Expected Outputs & Interview Questions

Based on your Day 6 assignment: personal profile page, student registration form, student records table, and semantic HTML elements.

This guide includes complete HTML code, explanations of the main tags, expected browser output, interview-ready explanations, and revision questions.

1. Introduction to HTML

HTML stands for HyperText Markup Language. It is a markup language used to define the structure and content of webpages. HTML can create headings, paragraphs, links, images, lists, tables, and forms.

HTML is not a programming language. CSS is used to style a page, while JavaScript adds interactivity and behavior.

Basic HTML document structure

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My First Page</title>
</head>
<body>
    <h1>Hello World</h1>
    <p>My first HTML page.</p>
</body>
</html>
```

Key tags: <!DOCTYPE html> declares HTML5; <html> is the root element; <head> contains metadata; <title> sets the browser-tab title; and <body> contains visible content.

2. Task 1 — Personal Profile Page

Objective: Display an introduction, education, skills, hobbies, and contact information using HTML.

Save this as **profile.html**. If you want a profile photo, place an image named **profile.jpg** in the same folder.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Personal Profile</title>
</head>
<body>
    <header>
        <h1>My Personal Profile</h1>
        <p>Welcome to my personal webpage!</p>
    </header>

    <nav>
        <a href="#about">About Me</a> |
        <a href="#education">Education</a> |
        <a href="#skills">Skills</a> |
        <a href="#contact">Contact</a>
    </nav>

    <main>
        <section id="about">
            <h2>About Me</h2>
            <img src="profile.jpg" alt="My Profile Photo" width="150">
            <p>Hello! My name is Shruti.</p>
            <p>I am a Computer Engineering graduate interested in learning new technologies and
building web applications.</p>
        </section>

        <section id="education">
            <h2>Education</h2>
            <ul>
                <li>B.E. in Computer Engineering</li>
                <li>Higher Secondary Education (12th)</li>
                <li>Secondary School Education (10th)</li>
            </ul>
        </section>

        <section id="skills">
            <h2>Technical Skills</h2>
            <ol>
                <li>HTML</li>
                <li>CSS</li>
```

```
                <li>Java</li>
                <li>MySQL</li>
            </ol>
        </section>

        <section id="hobbies">
            <h2>My Hobbies</h2>
            <ul>
                <li>Learning new technologies</li>
                <li>Reading</li>
                <li>Listening to music</li>
            </ul>
        </section>

        <section id="contact">
            <h2>Contact Information</h2>
            <p>Email: <a href="mailto:example@gmail.com">example@gmail.com</a></p>
            <p>Visit <a href="https://www.linkedin.com/" target="_blank" rel="noopener
noreferrer">LinkedIn</a></p>
        </section>
    </main>

    <footer>
        <p>&copy; 2026 My Personal Profile</p>
    </footer>
</body>
</html>
```

Important tags: **header** introduces the page; **nav** contains navigation links; **main** contains the main content; **section** groups related content; **img** displays an image; **a** creates a link; and **footer** contains footer information.

Expected output: A page heading, navigation links, profile section, education and skills lists, hobbies, contact links, and footer. The page uses the browser's default styling because no CSS is included.

Interview explanation: “I created a personal profile webpage using HTML. It displays my introduction, education, technical skills, hobbies, and contact information. I used headings, paragraphs, lists, links, and images, and organized the content with semantic elements.”

3. Task 2 — Student Registration Form

Objective: Collect student details using text fields, radio buttons, checkboxes, a dropdown, and a text area.

Save this as **registration.html**.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Registration Form</title>
</head>
<body>
    <header>
        <h1>Student Registration Form</h1>
        <p>Please fill in your details below.</p>
    </header>

    <main>
        <form action="#" method="post">
            <fieldset>
                <legend>Personal Information</legend>

                <label for="name">Full Name:</label>
                <input type="text" id="name" name="name"
                       placeholder="Enter your full name" required>
                <br><br>

                <label for="email">Email:</label>
                <input type="email" id="email" name="email"
                       placeholder="Enter your email" required>
                <br><br>

                <label for="phone">Phone Number:</label>
                <input type="tel" id="phone" name="phone"
                       placeholder="Enter your phone number" required>
                <br><br>
```

```
            <label for="dob">Date of Birth:</label>
            <input type="date" id="dob" name="dob" required>
            <br><br>

            <p>Gender:</p>
            <input type="radio" id="male" name="gender" value="Male">
            <label for="male">Male</label>

            <input type="radio" id="female" name="gender" value="Female">
            <label for="female">Female</label>

            <input type="radio" id="other" name="gender" value="Other">
            <label for="other">Other</label>
        </fieldset>

        <br>

        <fieldset>
            <legend>Educational Information</legend>

            <label for="course">Select Course:</label>
            <select id="course" name="course" required>
                <option value="">--Select Course--</option>
                <option value="BE">B.E.</option>
                <option value="BTech">B.Tech</option>
                <option value="BCA">BCA</option>
                <option value="MCA">MCA</option>
            </select>

            <br><br>

            <p>Choose Your Skills:</p>
            <input type="checkbox" id="html" name="skills" value="HTML">
            <label for="html">HTML</label>

            <input type="checkbox" id="java" name="skills" value="Java">
            <label for="java">Java</label>

            <input type="checkbox" id="mysql" name="skills" value="MySQL">
            <label for="mysql">MySQL</label>

            <br><br>

            <label for="address">Address:</label><br>
            <textarea id="address" name="address" rows="4" cols="35"
                      placeholder="Enter your address"></textarea>
        </fieldset>

        <br>
        <button type="submit">Register</button>
        <button type="reset">Reset</button>
    </form>
</main>

<footer>
    <p>Student Registration Portal</p>
</footer>
</body>
</html>
```

Key concepts:

• **form:** groups form controls. **label:** describes a field. **input:** accepts data. **select/option:** creates a dropdown. **textarea:** accepts multiple lines. **fieldset/legend:** groups related controls. **required:** asks the browser to require a value before normal submission.

• **Radio buttons** with the same name form a group where the user generally chooses one option. **Checkboxes** allow multiple independent selections.

• **id** identifies an element in the document; **name** identifies the field in submitted form data. A label's for value should match the input's id.

Expected output: A registration form with name, email, phone, date of birth, gender, course, skills, address, Register, and Reset controls.

Important limitation: This is an HTML demonstration. action="#" does not save records to a database. A backend or suitable form-processing service is needed to process and store submissions.

Interview explanation: “I created a student registration form using HTML form controls. It collects personal and educational information using inputs, radio buttons, checkboxes, a dropdown, and a text area. I used labels and required fields for usability and basic browser validation.”

4. Task 3 — Student Records Table

Objective: Display student records in rows and columns.

Save this as **students.html**.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Records</title>
</head>
<body>
    <header>
        <h1>Student Records</h1>
        <p>Details of registered students.</p>
    </header>

    <main>
        <section>
            <h2>Student Information Table</h2>
            <table border="1">
                <caption>Student Academic Records</caption>

                <thead>
                    <tr>
                        <th>Student ID</th>
                        <th>Name</th>
                        <th>Course</th>
                        <th>Marks</th>
                    </tr>
                </thead>

                <tbody>
                    <tr><td>101</td><td>Amit</td><td>B.E. Computer Engineering</td><td>85</td></tr>
                    <tr><td>102</td><td>Priya</td><td>B.E. Computer Engineering</td><td>92</td></tr>
                    <tr><td>103</td><td>Rahul</td><td>B.Tech</td><td>78</td></tr>
                    <tr><td>104</td><td>Neha</td><td>BCA</td><td>88</td></tr>
                </tbody>

                <tfoot>
                    <tr>
                        <td colspan="3">Total Students</td>
                        <td>4</td>
                    </tr>
                </tfoot>
            </table>
        </section>
    </main>

    <footer>
        <p>End of Student Records</p>
    </footer>
</body>
</html>
```

Table tags:

• **table:** creates a table. **caption:** gives it a title. **thead:** groups header rows. **tbody:** groups data rows. **tfoot:** groups footer rows. **tr:** creates a row. **th:** creates a header cell. **td:** creates a data cell.

colspan="3" makes a cell span three columns. The border attribute is a legacy presentation method; CSS is preferred for modern table styling.

Expected output: A four-column table showing student ID, name, course, and marks, followed by a total of four students.

Interview explanation: “I created a student records table using HTML. I used table rows, header cells, data cells, and thead/tbody/tfoot to organize the records. I used colspan to display the total-student label across three columns.”

5. Task 4 — Semantic HTML

Objective: Structure a webpage using elements that describe the meaning of each content area.

Save this as **semantic.html**.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Semantic HTML Example</title>
</head>
<body>
    <header>
        <h1>My Learning Website</h1>
        <p>Learn HTML fundamentals.</p>
    </header>

    <nav>
        <a href="#home">Home</a> |
        <a href="#courses">Courses</a> |
        <a href="#contact">Contact</a>
    </nav>

    <main id="home">
        <section id="courses">
            <h2>Our Courses</h2>

            <article>
                <h3>HTML Fundamentals</h3>
                <p>Learn the basic structure of webpages.</p>
            </article>

            <article>
                <h3>CSS Fundamentals</h3>
                <p>Learn how to style webpages.</p>
            </article>
        </section>

        <aside>
            <h2>Learning Tip</h2>
            <p>Practice HTML by creating small webpages.</p>
        </aside>
    </main>

    <footer id="contact">
        <p>Contact: example@gmail.com</p>
        <p>&copy; 2026 My Learning Website</p>
    </footer>
</body>
</html>
```

Semantic elements:

• **header:** introductory content. **nav:** navigation links. **main:** primary page content. **section:** related thematic content. **article:** self-contained content. **aside:** complementary information. **footer:** footer details.

A **div** is a generic container with no inherent semantic meaning. A **section** groups related content, usually under a heading. An **article** is suitable for self-contained content, such as a blog post or course description.

Benefits: Semantic HTML can improve code readability and maintainability, support accessibility, and help search engines understand page structure.

Interview explanation: “I used semantic HTML elements to organize content by purpose. This makes the webpage structure easier to understand and maintain and can improve accessibility.”

6. Important HTML Interview Questions

1. What is HTML?

HTML stands for HyperText Markup Language. It defines the structure and content of webpages.

2. What is HTML5?

HTML5 refers to the major HTML revision that introduced semantic elements and other features used by modern webpages.

3. What is an HTML element?

An element generally includes a start tag, content, and an end tag, for example Hello . Some elements, such as img and br, are void elements.

4. What is an attribute?

An attribute provides extra information or configuration for an element, for example href in a link.

5. What are heading tags?

HTML provides six heading levels, h1 through h6, with h1 the highest-level heading and h6 the lowest.

6. Difference between HTML and CSS?

HTML defines structure and content; CSS controls appearance and layout.

7. Difference between HTML and JavaScript?

HTML defines content and structure; JavaScript adds behavior and interactivity.

8. What is a hyperlink?

A link created with the a element, usually using the href attribute.

9. What are block and inline elements?

Block elements normally start on a new line and take available width; inline elements normally flow within text. CSS can change display behavior.

10. What does br do?

It creates a line break.

11. What does hr do?

It represents a thematic break, commonly displayed as a horizontal line.

12. Difference between id and class?

An id should uniquely identify an element in the document; a class can be shared by multiple elements.

13. What does target="_blank" do?

It requests opening a link in a new browsing context, commonly a new tab. rel="noopener noreferrer" is a useful accompanying security/privacy measure.

14. Can HTML save data to a database?

No. HTML creates the form; backend code or a suitable service must process and store submitted data.

15. Difference between th and td?

th defines a table header cell; td defines a regular data cell.

16. What does the form action attribute do?

It specifies the destination that receives a form submission.

17. Difference between GET and POST?

GET commonly retrieves information and places query parameters in the URL. POST commonly submits data in the request body. POST alone does not provide security; HTTPS is important.

18. Difference between head and header?

head contains document metadata; header is a semantic element for introductory content in the page.

7. How to Run the Programs

1. Create a folder named **Day6_HTML**.
2. Create four files: profile.html, registration.html, students.html, and semantic.html.
3. Copy the relevant program into each file and save it with the .html extension.
4. Open each file in a browser or use your code editor's preview feature.
5. Test the links, form controls, table rows, and semantic page structure.

Profile page: Confirm the links work and the image path is correct. **Registration form:** Try required fields, radio buttons, checkboxes, and Reset. **Student table:** Check all four records and the total. **Semantic page:** Check the navigation and content groups.

8. Final Revision Checklist

Topic	Remember
HTML	Defines webpage structure and content.
Document structure	html, head, body
Headings and paragraphs	h1–h6, p
Links and images	a with href; img with src and alt
Lists	ul, ol, li
Tables	table, tr, th, td
Forms	form, label, input, select, textarea, button
Radio vs checkbox	Radio: one choice in a group; checkbox: multiple choices.
Semantic HTML	header, nav, main, section, article, aside, footer
CSS	Controls appearance and layout.
JavaScript	Adds interactivity and behavior.
Backend	Processes submissions and can store data in a database.

You have now covered the four Day 6 assignment tasks: personal profile webpage, student registration form, student records table, and semantic HTML.